Number Representation and Arithmetic Circuits:

Signed Numbers, Binary Adders and Subtractors

Octal and hexadecimal numbers

- Positional notation can be used for any radix (base). If the radix is *r*, then the number

  $K=k_{n-1}k_{n-2}. \; . \; .k_1k_0$

  has the value

$$V(K)=\sum_{i=0}^{n-1} k_i \times r^i$$

- Numbers with radix-8 are called ***octal*** and numbers with radix-16 are called ***hexadecimal*** (or hex)
  - For octal, digit values range from 0 to 7
  - For hex, digital values range from 0-9 and A-F

Numbers in different systems

Decimal	Binary	Octal	Hex
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7

Decimal	Binary	Octal	Hex
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Binary to hex or octal conversion

- **Group binary digits into groups of four and assign each group a hexadecimal digit.**

0110	1011	0111
6	B	7

  - **Binary-to-octal:**

011	010	110	111
3	2	6	7

  - **Hexadecimal-to-binary:**

A	1	9
1010	0001	1001

  - **Octal-to-binary:**

5	0	3	1
101	000	011	001

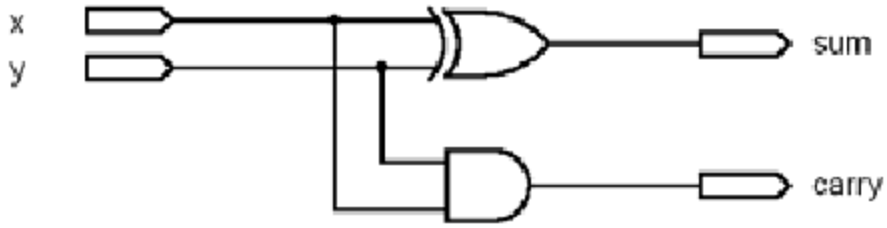
Unsigned number addition

- Additional of two 1-bit numbers gives four possible combinations

x	0	0	1	1
+ y	+ 0	+ 1	+ 0	+ 1
c s	0 0	0 1	0 1	1 0

carry sum

x	*y*	*c*	*s*
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0



Unsigned number addition

- Larger numbers have more bits involved
  - There is still the need to add each pair of bits
  - But, for each bit position *i*, the addition operation may include a ***carry-in*** from bit position *i-1*

$X=x_4x_3x_2x_1x_0$	0 1 1 1 1	$(15)_{10}$
$Y=y_4y_3y_2y_1y_0$	0 1 0 1 0	$(10)_{10}$
	1 1 1 0	Generated carries
$S=s_4s_3s_2s_1s_0$	1 1 0 0 1	$(25)_{10}$

Full adder circuit

c_i	x_i	y_i	c_{i+1}	s_{i+1}
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

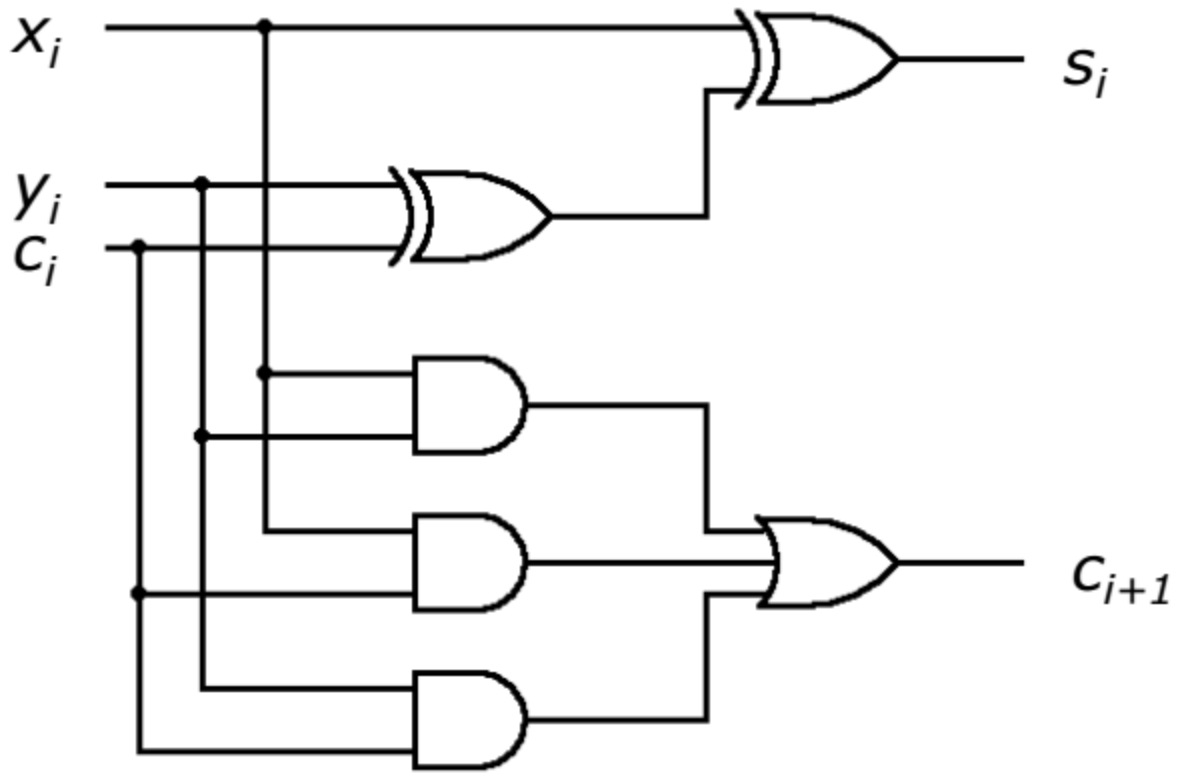
c_i \ x_iy_i	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$s_i = x_i \oplus y_i \oplus c_i$$

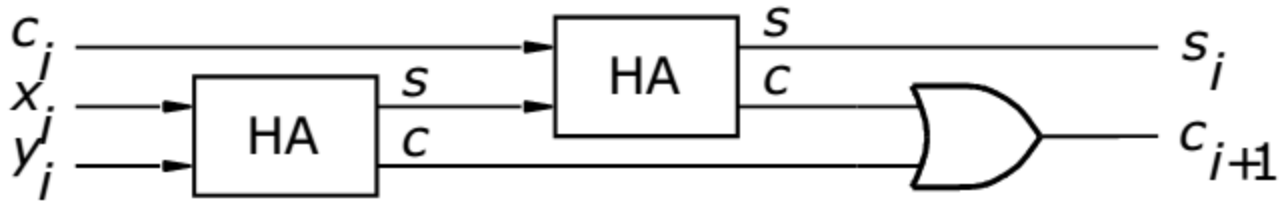
c_i \ x_iy_i	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$c_i = x_i y_i + y_i c_i + x_i c_i$$

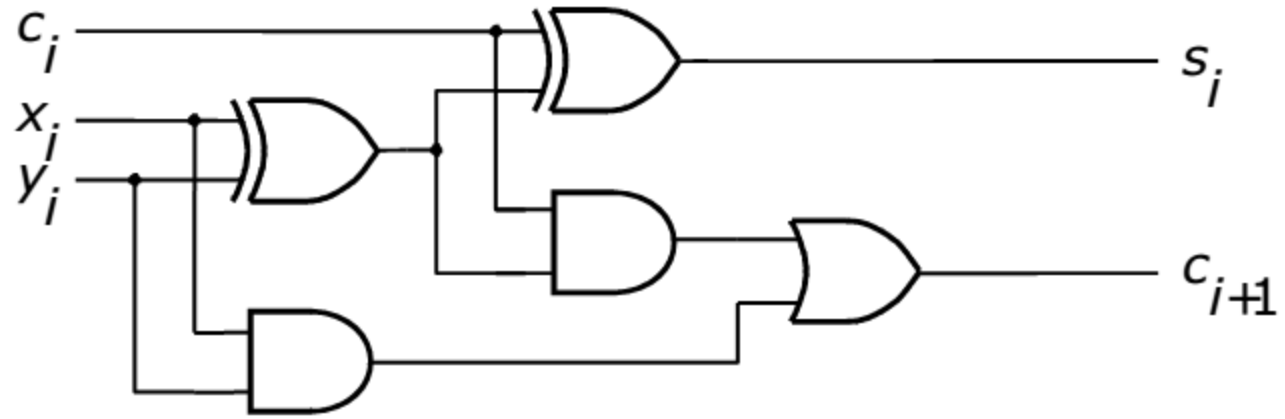
Full adder circuit



Full adder circuit (decomposed)



Block diagram

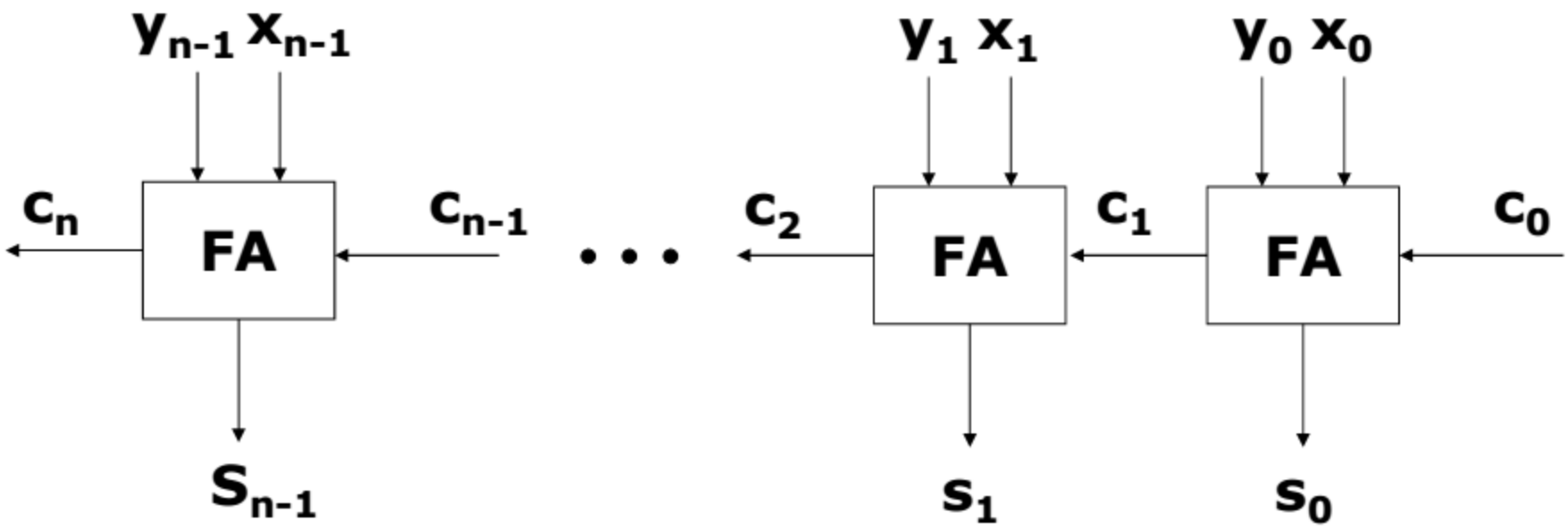


Detailed diagram

Ripple-carry adder

- In performing addition, we start from the least significant digit and add pairs of digits progressing to the most significant digit
- If a carry is produced in position *i*, it is added to operands (digits) in position *i*+1
- A chain of full adders, connected in sequence, can perform this operation
- Such a configuration is called a ***ripple-carry adder*** because of the way the carry signal 'ripple' through from stage to stage

Ripple-carry adder

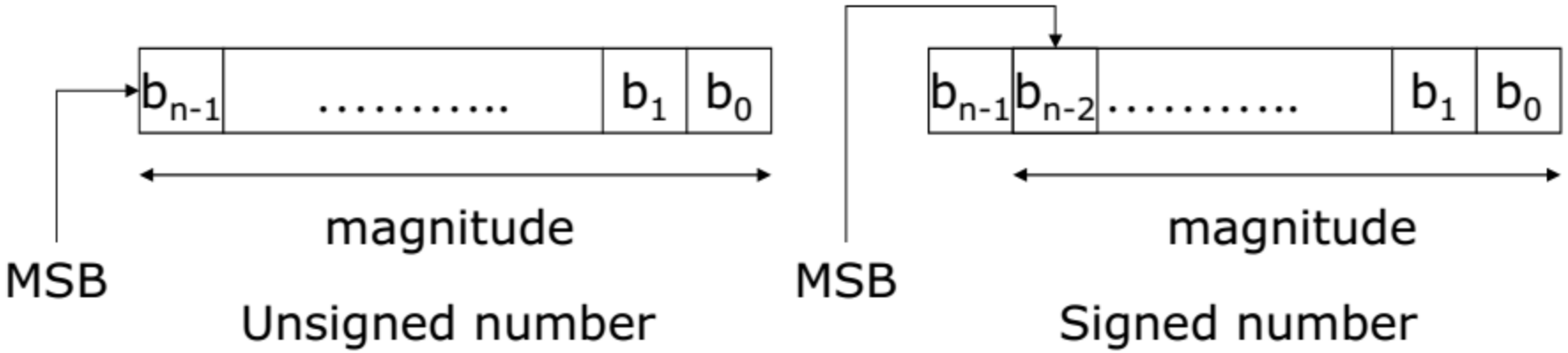


Ripple-carry adder

- Each full adder introduces a certain delay before its s_i and c_{i+1} outputs are valid
  - The propagation delay through the full adder
- Let this delay be Δt
- The carry out of the first stage c_1 arrives at the second stage Δt after the application of the x_0 and y_0 inputs
- The carry out of the second stage c_2 arrives at the third stage with a delay of $2\Delta t$, and so on
- The signal c_{n-1} is valid after $(n-1)\Delta t$, and the complete sum is available after a delay of $(n)\Delta t$
- The delay obviously depends on the size of the numbers (*i.e.* the number of bits)

Signed numbers

- For signed numbers, in the binary system, the sign of the number is denoted by the left-most bit
  - 0 = positive
  - 1 = negative
- For an n-bit number, the remaining *n*-1 bits represent the magnitude



Negative numbers

- For signed numbers, there are three common formats for representing negative numbers
  - Sign-magnitude
  - 1's complement
  - 2's complement
- Sign-magnitude uses one bit for the sign (0=+, 1=-) and the remaining bits represent the magnitude of the number as in the case of unsigned numbers
- For example, using 4-bit numbers

  +5=0101 -5=1101

  +3=0011 -3=1011

  +7=0111 -7=1111
- Although this is easy to understand, it is not well suited for use in computers

1's complement representation

- In the 1's complement scheme, an *n*-bit negative number *K*, is obtained by subtracting its equivalent positive number, *P*, from 2^n-1

  $K=(2^n-1)-P$

- For example, if n=4, then

  $K=(2^4-1)-P=(15)_{10}-P=(1111)_2-P$

  $-5=(15)_{10}-5=(1111)_2-(0101)_2=(1010)_2$

  $-3=(15)_{10}-3=(1111)_2-(0011)_2=(1100)_2$

- From these examples, clearly the 1's complement can be formed simply by complementing each bit of the number, including the sign bit
- Numbers in the 1's complement form have some drawbacks when used in arithmetic operations

2's complement representation

- In the 2's complement scheme, an *n*-bit negative number *K*, is obtained by subtracting its equivalent positive number, *P*, from 2^n

  $K=2^n-P$

- For example, if n=4, then

  $K=2^4-P=(16)_{10}-P=(10000)_2-P$

  $-5=(16)_{10}-5=(10000)_2-(0101)_2=(1011)_2$

  $-3=(16)_{10}-3=(10000)_2-(0011)_2=(1101)_2$

- A simple way of finding the 2's complement of a number is to add 1 to its 1's complement

Rule for finding 2's complements

- Given a signed number, $B=b_{n-1}b_{n-2}\ldots b_1b_0$, its 2's complement, $K=k_{n-1}k_{n-2}\ldots k_1k_0$, can be found by:
  - examining all the bits of *B* from right to left and complementing all the bits after the first '1' is encountered
- For example if

  B=00110100
- Then the 2's complement of B is

  K=11001100

changed bits unchanged bits

Four-bit signed integers

$b_3b_2b_1b_0$	Sign-magnitude	1's complement	2's complement
0111	+7	+7	+7
0110	+6	+6	+6
0101	+5	+5	+5
0100	+4	+4	+4
0011	+3	+3	+3
0010	+2	+2	+2
0001	+1	+1	+1
0000	+0	+0	+0
1000	-0	-7	-8
1001	-1	-6	-7
1010	-2	-5	-6
1011	-3	-4	-5
1100	-4	-3	-4
1101	-5	-2	-3
1110	-6	-1	-2
1111	-7	-0	-1

Addition and subtraction

- For sign-magnitude numbers, addition is simple, but if the numbers have different signs the task becomes more complicated
  - Logic circuits that compare and subtract numbers are also needed
  - It is possible to perform subtraction without this circuitry
  - For this reason, sign-magnitude is not used in computers
- For 1's complement numbers, adding or subtracting some numbers may require a correction to obtain the actual binary result
- For example, (-5)+(-2)=(-7), but when adding the binary equivalents of –5 and –2, the result is 0111 with an additional carry out of 1 which must be added back the the result to produce the final (correct) result of 1000

2's complement operations

- For addition, the result is always correct
- Any carry-out from the sign-bit position is simply ignored

(+5)	0 1 0 1	(–5)	1 0 1 1
+(+2)	+0 0 1 0	+(+2)	+0 0 1 0
(+7)	0 1 1 1	(–3)	1 1 0 1
(+5)	0 1 0 1	(–5)	1 0 1 1
+ (–2)	+1 1 1 0	+ (–2)	+1 1 1 0
(+3)	1 0 0 1 1	(–7)	1 1 0 0 1

ignore

ignore

2's complement subtraction

- The easiest way of performing subtraction is to negate the subtrahend and add it to the minuend
  - Find the 2's complement of the subtrahend and then perform addition

(+5)	0 1 0 1	⇒	0 1 0 1
−(+2)	− 0 0 1 0		+ 1 1 1 0
(+3)			1 0 0 1 1

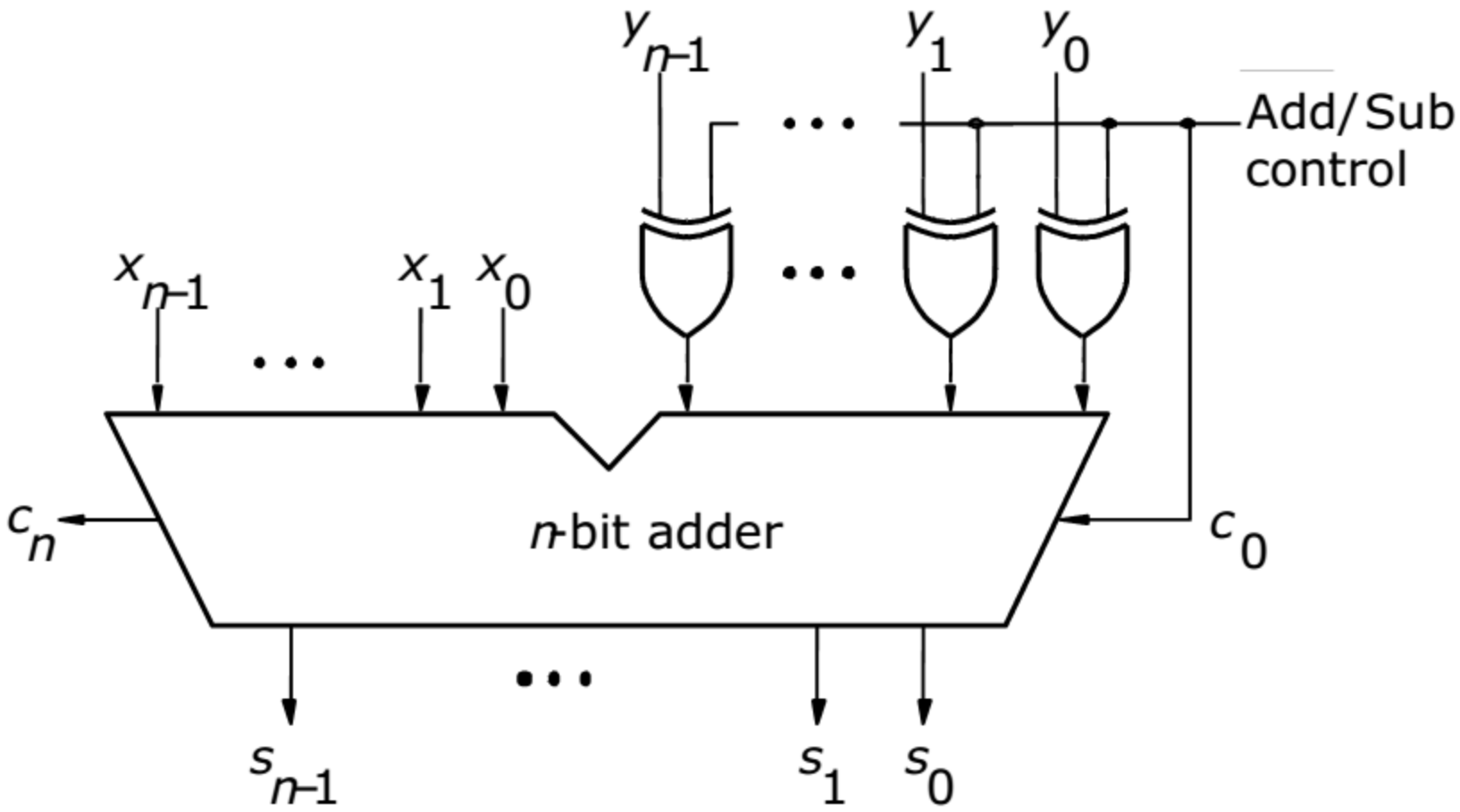
ignore

(- 5)	1 0 1 1	⇒	1 0 1 1
−(- 2)	− 1 1 1 0		+ 0 0 1 0
(- 3)			1 1 0 1

Adder and subtractor unit

- The subtraction operation can be realized as the addition operation, using a 2's complement of the subtrahend, regardless of the signs of the two operands
  - It is possible to use the same adder circuit to perform both addition and subtraction
- Recall that the 2's complement can be formed from the 1's complement simply by adding 1
- We can use the XOR operation to perform a 1's complement
  - Recall $x \oplus 1 = x'$ and $x \oplus 0 = x$
  - If we are performing a subtract operation, 1's complement the subtrahend by XORing each bit with 1

Adder and subtractor unit



Arithmetic overflow

- The result of addition or subtraction is supposed to fit within the significant bits used to represent the numbers
- If *n* bits are used to represent signed numbers, then the result must be in the range -2^{n-1} to $+2^{n-1}-1$
- If the result does not fit in this range, we say that ***arithmetic overflow*** has occurred
- To insure correct operation of an arithmetic circuit, it is important to be able to detect the occurrence of overflow

Examples for determining arithmetic overflow

For 4-bit numbers, there are 3 significant bits and the sign bit

(+7)	0 1 1 1	(–7)	1 0 0 1
+(+2)	+0 0 1 0	+(+2)	+0 0 1 0
(+9)	1 0 0 1	(–5)	1 0 1 1
	$c_4 = 0$		$c_4 = 0$
	$c_3 = 1$		$c_3 = 0$

(+7)	0 1 1 1	(–7)	1 0 0 1
+(–2)	+1 1 1 0	+(–2)	+1 1 1 0
(+5)	1 0 1 0 1	(–9)	1 0 1 1 1
	$c_4 = 1$		$c_4 = 1$
	$c_3 = 1$		$c_3 = 0$

If the numbers have different signs, no overflow can occur

Arithmetic overflow

- In the previous examples, overflow was detected by

  $overflow = c_3 c_4' + c_3' c_4$

  $overflow = c_3 \oplus c_4$

- For n-bit numbers we have

  $overflow = c_{n-1} \oplus c_n$

- The adder/subtractor circuit introduced can be modified to include overflow checking with the addition of one XOR gate